Topic 4: Multiple Regression Analysis

ECON 4753 — University of Arkansas

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Outline

Multiple Variable Regression

Polynomials

Interactions

Regressions compare people

A bivariate regression compares people with different values of X to describe how their Y differs.

- For example, a regression of wages on years of education compares people with more education to people with less education.

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- For example, a regression of wages on years of education compares people with more education to people with less education.

But people with different values of X can differ in many other ways!

- People with more education may also have different family backgrounds, work experience, and skills.
- These other differences can help explain the wage differences we see.

Other differences in the comparison

Wine and health



Suppose we compare wine drinkers to people who do not drink wine and find that wine drinkers are healthier.

Other differences in the comparison

Wine and health



Suppose we compare wine drinkers to people who do not drink wine and find that wine drinkers are healthier.

What else might differ between these groups?

- Income, education, diet, and access to health care.
- These things affect health too!

Omitted variables

An omitted variable is a variable left out of our regression.

An omitted variable can help explain the relationship between X and Y when it:

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2. Affects Y beyond the relationship with X.

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For wine and health, income could satisfy both conditions.

- If wine drinkers have higher incomes, our comparison also compares people with different incomes.
- Income can affect health through diet, housing, and access to health care.

Examples of Omitted Variables

X **and** Y	**What else might explain the comparison?**
Education and wages	Family resources can help people stay in school and find higher-paying jobs.
Tutoring and test scores	Students who struggle may seek more tutoring, and prior preparation affects test scores.
Hospital care and health	Sicker people are more likely to seek hospital care, and illness severity affects health outcomes.

Examples of Omitted Variables

Compare X and Y	**What else might explain the comparison?**
House size and price	Larger houses may be in different neighborhoods, and location affects price.
Arts spending and test scores	Wealthier districts may spend more on arts and provide other resources that improve test scores.

Control variables

Multiple regression lets us include these other variables as control variables.

- We will call the variable of interest X and the control variables W.
- W can contain one control or several controls, $W_1, \ldots, W_J$.

Comparing people with the same W

When using control variables, the way we interpret the regression is going to change:

Compare two people with the same values of the control variables, W, but different values of X. How do their predicted values of Y differ?

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> *Compare two people with the same values of the control variables, W, but different values of X. How do their predicted values of Y differ?*

We describe the relationship between X and Y while *holding the control variables W fixed*.

Comparing people with the same W

For example, compare two people with the same family income and years of work experience, whose education differs by one year.

- Y is wages, X is years of education, and W contains family income and work experience.
- We call this the relationship between education and wages *controlling for* family income and work experience.

Multiple regression

With one control variable W_i, our model is

$$y_i = \beta_0 + \beta_1 X_i + \gamma W_i + u_i.$$

Compare two people with the same W whose X differs by one unit. Their predicted outcomes differ by $\hat{\beta}_1$:

$$\hat{y}_j - \hat{y}_i = \hat{\beta}_1 \underbrace{(X_j - X_i)}_{1} + \hat{\gamma} \underbrace{(W_j - W_i)}_{0} = \hat{\beta}_1.$$

Multiple regression

Several controls

With J control variables, we write:

$$y_i = \beta_0 + \beta_1 X_{1,i} + \cdots + \beta_K X_{K,i} + u_i.$$

β_1 describes the relationship between X_1 and Y, holding *all the other included controls* fixed.

Advertising example

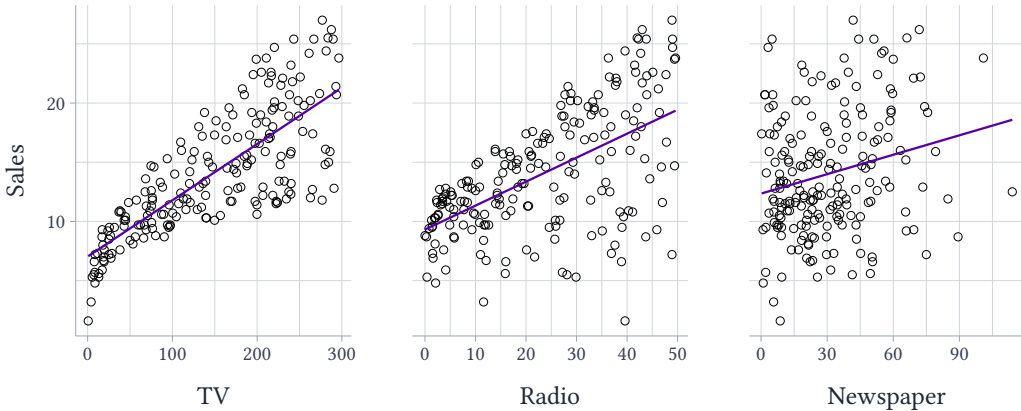
Suppose we have data on sales and spending on TV, radio, and newspaper ads across different markets.

We want to describe the relationship between radio advertising and sales.

- Y is sales and X is spending on radio ads.
- Our controls W are spending on TV and newspaper ads.

Advertising example

Single-variable predictors



Advertising example

What else differs between markets?

Each plot compares markets with different spending on one type of advertising.

Advertising example

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Each plot compares markets with different spending on one type of advertising.

Markets with more radio ads might also spend more on TV and newspaper ads.

- The bivariate regression compares markets that can differ in all of these ways.
- Its radio coefficient can reflect differences in other advertising spending too.

Advertising example

Holding W fixed

Compare two markets with the same spending on TV and newspaper ads, but different spending on radio ads.

Our multiple regression is

$$\text{Sales}_i = \beta_0 + \beta_1 \text{Radio}_i + \gamma_1 \text{TV}_i + \gamma_2 \text{Newspaper}_i + u_i.$$

Advertising example

Holding W fixed

Compare two markets with the same spending on TV and newspaper ads, but different spending on radio ads.

Our multiple regression is

$$\text{Sales}_i = \beta_0 + \beta_1 \text{Radio}_i + \gamma_1 \text{TV}_i + \gamma_2 \text{Newspaper}_i + u_i.$$

$\hat{\beta}_1$ is the predicted difference in sales for one more unit of radio spending, *holding TV and newspaper spending fixed*.

Dependent Variable:	Sales			
Constant	7.033***	9.312***	12.35***	2.939***
	(0.4578)	(0.3882)	(0.6338)	(0.3365)
TV	0.0475***			0.0458***
	(0.0027)			(0.0019)
Radio		0.2025***		0.1885***
		(0.0217)		(0.0108)
Newspaper			0.0547***	-0.0010
			(0.0186)	(0.0064)
R^2	0.61188	0.33203	0.05212	0.89721

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Interpreting regression results

In the bivariate regression, higher newspaper advertising spending is associated with higher sales.

After controlling for TV and radio spending, the estimated newspaper coefficient is close to zero.

- Comparing markets with the same TV and radio spending gives a different relationship than the bivariate comparison.
- The results suggest that differences in TV and radio spending help explain the bivariate relationship between newspaper ads and sales.

Controls and causal interpretation

Suppose we regress wages on college attendance, controlling for family income and SAT scores.

We compare people with the same values of these controls who differ in college attendance.

- They may still differ in motivation, professional connections, or other determinants of wages.
- If these omitted variables are correlated with college attendance and affect wages, they can still help explain the comparison.

Controls and causal interpretation

Multiple regression holds the *included controls* W fixed.

A causal interpretation requires that no remaining differences related to X also affect Y. Adding controls does not automatically establish this.

Outline

Multiple Variable Regression

Inference on Regression Coefficients

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Inference on regression coefficients

Conducting inference on regression coefficients is the same as before

- We get point estimates and standard errors just as before

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- We get point estimates and standard errors just as before

However, the standard errors are going to depend on other covariates.

- Whenever the covariate of interest is highly correlated with other variables, the standard errors on our estimate will grow
  - $\rightarrow$ there is not much independent variation left after controlling for the other variables (so our estimates are noisier)

Multicollinearity and Standard Errors

Consider a regression of SAT scores among middle and high-school students. Imagine regressing SAT score on the age of the student and the student's grade

Age and a student's grade are highly correlated, so it's very hard to distinguish between the two

$\Longrightarrow$ the standard errors on each will be very large

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Quadratic terms

Say you have the following model with wages as a quadratic function of age

$$w_i = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{age}_i^2 + u_i$$

Before we were discussing the idea of changing one variable, while holding the rest "equal"

- How does one change age without changing age^2?

Marginal Effects

Really what we were doing was writing out the predicted value as

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1,i} + \cdots + \hat{\beta_K} X_{K,i} + u_i$$

Then the marginal effect of the ℓ-th variable was the change in $\hat{y}_i$ when changing $X_{\ell,i}$ but holding the rest of the variables *fixed*

- This is exactly what the partial-derivative tells us

Marginal Effects

That is, our marginal effect was $\frac{\partial \hat{y}_i}{\partial X_{\ell,i}}$ evaluated at the original covariate values: $X_{1,i}, \ldots, X_{K,i}$.

Marginal effects with quadratic terms

$$\hat{w}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{age}_i + \hat{\beta}_2 \text{age}_i^2$$

From calculus, we know that the partial derivative of $\hat{w}_i$ with respect to age_i is given by

$$\frac{\partial \hat{w}_i}{\partial \text{age}_i} = \hat{\beta}_1 + 2\hat{\beta}_2 \text{age}_i$$

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$$\frac{\partial \hat{w}_i}{\partial \text{age}_i} = \hat{\beta}_1 + 2\hat{\beta}_2 \text{age}_i$$

The change in predicted wage of a worker as they age is given by $\hat{\beta}_1 + 2\hat{\beta}_2 \text{age}_i$ which depends on their age

- In words, how much predicted wages change as a worker gets a year older changes over a worker's lifetime

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Why interactions

Wages Example

Consider a model where we want to understand how wages are influenced by both being a female and being a college-educated worker. We can write the model as:

$$w_i = \beta_0 + \beta_1 \text{female}_i + \beta_2 \text{college}_i + \beta_3 (\text{female}_i \times \text{college}_i) + u_i$$

Here, β_3 captures the interaction effect between female and college-education status on wages

- This means that the effect of females on wages may differ depending on whether the worker has a college-degree, and vice versa.

Interactions

Wages Example

Consider the difference in predicted wages for non-college educated male vs. non-college educated workers:

$$w_{i,NC,F} - w_{i,NC,M} = (\beta_0 + \beta_1) - \beta_0 = \beta_1$$

Compare this to the difference in predicted wages for college educated male vs. college educated workers:

$$w_{i,C,F} - w_{i,C,M} = (\beta_0 + \beta_1 + \beta_2 + \beta_3) - (\beta_0 + \beta_2) = \beta_1 + \beta_3$$

Interactions

Wages Example

Wage-gap for college educated workers is $\beta_1 + \beta_3$ and the wage-wagp for non-college educated workers is β_1

- β_3 represents the difference in wage gaps of college-educated vs. non-college-educated workers.

Interactions

Wages Example

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- β_3 represents the difference in wage gaps of college-educated vs. non-college-educated workers.

More generally, can interpret interactions how one variable changes the marginal effect of another variable

- This is similar to when you have a quadratic function of X, the marginal effect depends where you are along the distribution of X.

Interactions

Partial Derivative

$$\hat{w}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{female}_i + \hat{\beta}_2 \text{college}_i + \hat{\beta}_3 (\text{female}_i \times \text{college}_i)$$

We can derive this result using partial derivatives:

$$\frac{\partial \hat{w}_i}{\partial \text{female}_i} = \hat{\beta}_1 + \hat{\beta}_3 \text{college}_i$$

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$$\frac{\partial \hat{w}_i}{\partial \text{female}_i} = \hat{\beta}_1 + \hat{\beta}_3 \text{college}_i$$

In this case, it's a little weird to think of "small" changes in the female variable. Instead, we will think of it as a 1 unit change (from 0 to 1)

Continuous interacted with a discrete variable

Let X_i be a continuous variable and D_i be a dummy variable and consider the regression

$$y_i = \beta_0 + D_i\beta_1 + X_i\beta_2 + X_iD_i\beta_3 + u_i$$

Continuous interacted with a discrete variable

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$$y_i = \beta_0 + D_i\beta_1 + X_i\beta_2 + X_iD_i\beta_3 + u_i$$

In this case, the marginal effect of X_i is given by $\frac{\partial \hat{y}_i}{\partial X_i} = \hat{\beta}_2 + D_i\hat{\beta}_3$

- The marginal effect for group $D_i = 0$ is given by $\hat{\beta}_2$
- The marginal effect for group $D_i = 1$ is given by $\hat{\beta}_2 + \hat{\beta}_3$

Continuous interacted with a discrete variable

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In this case, the marginal effect of X_i is given by $\frac{\partial \hat{y}_i}{\partial X_i} = \hat{\beta}_2 + D_i\hat{\beta}_3$

- The marginal effect for group $D_i = 0$ is given by $\hat{\beta}_2$
- The marginal effect for group $D_i = 1$ is given by $\hat{\beta}_2 + \hat{\beta}_3$
- Therefore, $\hat{\beta}_3$ is the difference in marginal effects between $D_i = 1$ relative to $D_i = 0$

Continuous interacted with a discrete variable

$$y_i = \beta_0 + D_i\beta_1 + X_i\beta_2 + X_iD_i\beta_3 + u_i$$

Exercise:

- In words, how would you interpret a t-test for the null that $\hat{\beta}_2 = 0$?
- In words, how would you interpret a t-test for the null that $\hat{\beta}_3 = 0$?

`mtcars` example

```
OLS estimation, Dep. Var.: mpg
              Estimate Std. Error   t value   Pr(>|t|)
(Intercept) 26.624848   1.346754 19.769644  < 2.2e-16 ***
am::1        5.217653   2.324898  2.244250 3.2904e-02 *
hp          -0.059137   0.008957 -6.602265 3.6781e-07 ***
am::1:hp     0.000403   0.013362  0.030152 9.7616e-01
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Exercise: Does the estimate relationship between being a car's horsepower and miles per gallon depend on whether it is an automatic? How do you know?

Interaction terms always should have ‘main effects’

When including an interaction term, it is important to (almost) always include the main effects

The main effects are the variables by themselves. E.g. if you interact gender with race, you want to include race and gender as separate terms as well

- The main effects are what let us interpret the interaction term as the ‘difference’ in marginal effects